

Alternate Solutions *by Michael A. Gottlieb*

Here are three more approaches to solving the machine design problem presented earlier in this chapter (Section 2-7), beginning on p. 73.

A Finding the acceleration of the weight using geometry

The weight is always horizontally halfway between the roller and the pivot, so its horizontal speed is 1 m/s, half the speed of the roller. The weight moves on a circle (centered at the pivot), so its velocity is perpendicular to the rod. By similar triangles we obtain the velocity of the weight. (See Fig. 2-14a.)

Because the weight moves on a circle, the radial component of its acceleration is

$$a_{\text{rad}} = \frac{v^2}{r} = \frac{(1.25)^2}{0.5} = 3.125.$$

as per Eq. (2.17). The vertical acceleration of the weight is the sum of its radial and transverse components. (See Fig. 2-14b.)

Using similar triangles again, we obtain the vertical acceleration:

$$a_y = \frac{a_y}{a_{\text{rad}}} \times a_{\text{rad}} = \frac{0.5}{0.4} \times 3.125 = 3.90625.$$

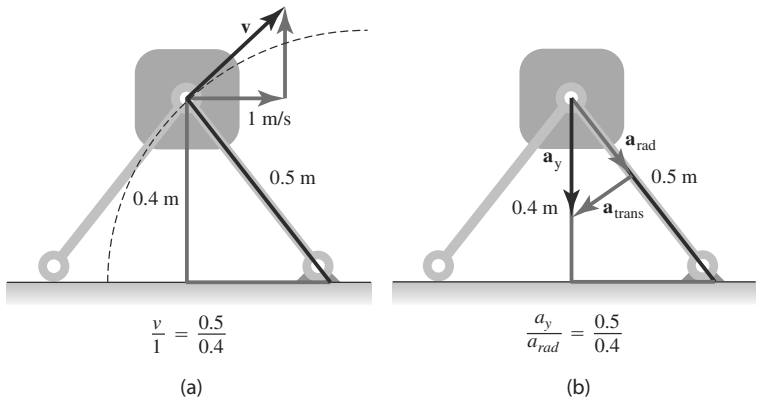


FIGURE 2-14